Fairness for me but not for thee: The group perspective on fairness in multiwinner elections

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Abstract

Designing metrics of fairness faces some common structural challenges across application domains, and a principal one is which group's perspective is taken in the measurement. We introduce the perspective question into the domain of social choice theory, or the mathematical study of electoral systems and decision rules. In computational social choice, "metric voting" evaluates voting rules by their ability to elect a lowest-cost candidate or winner set, measured by distance in a latent metric space where both candidates and voters are embedded according to ideological positions. The standard evaluations are group-neutral: they sum cost over all voters or look at the worst-off subset, without regard to identifying features, thereby treating voters as interchangeable. We propose a new definition of group inefficiency of a voting rule, which measures cost from the standpoint of a group of voters with some salient shared identity or views. Simulated elections show that this shift in perspective drastically changes which multi-winner voting rules look fair. To back this up, we provide rigorous results, both in Euclidean and more general metric settings, including some with asymptotic guarantees. Finally, a sweep of over 700 real-world elections grounds the work in reality and shows that even in noisy observed conditions, the structural findings hold up. This study should be read against the tendency in the fairness literature to take protected groups as given, or to demand fairness for every definable subgroup; instead, we argue that the choice of group perspective is itself a normative commitment.

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1 Introduction

Across application domains, questions of fairness ask whether the interests of individuals and of population subgroups are adequately represented when the system is evaluated in aggregate [24, 20]. A recurring lesson is that aggregate measures can look acceptable while a specific, recognizable group is poorly served [7], so which groups to center is itself a substantive choice [31, 22]. The same question arises in computational social choice (COMSOC), which studies collective decisions—including voting rules for democratic representation—using methods from computer science and economics, and where work to operationalize standards


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of proportional representation is on the upswing (as in [6] and references). However, there is a persistent gap between how subgroups are conceived in COMSOC and how they are defined by practitioners in democracy and voting rights spheres. The split comes down to which groups are centered: COMSOC standards typically must hold for every group of voters of a given size or cohesion, while practitioners start from particular cognizable groups with imperfectly shared voting patterns.

We bring the practitioner view together with the metric framework of computational social choice. There, a voting rule is evaluated by the *cost* of its winner set, measured through distances in a latent space of voters and candidates, relative to the best available alternative. Those kinds of cost ratio are usually computed over the whole electorate or over the worst-off group of a given size. We instead compute cost ratios from the perspective of designated groups of voters. This change of perspective does not just update, but even reverses some findings. Rules that favor broadly acceptable compromise candidates, like the Borda count, look nearly optimal from the whole-electorate view but are among the worst for each group. In sharp contrast, the group perspective sees these as weak options and instead favors rules like STV that split the winner set to provide highly preferred representation for each bloc—and even separates rules designed for proportionality into stronger and weaker performers. Worst-case definitions miss this dynamic, not only because the worst-off subset is rarely a group anyone would name, but also because worst-case measurements lose some of the signal that distinguishes the rules.

The group inefficiency ratio requires specifying a group or voting bloc. In our synthetic experiments, the voters are generated by a mixture model and the components are the blocs; in the Scottish data, blocs are found by clustering the ballots. In practice, a group might instead be designated by law—for instance, the U.S. Voting Rights Act of 1965 identifies racial, ethnic, or language groups as being covered by a distinctive set of protections [28]—or by an exogenous label such as party registration. Which groups should count is a question we return to in Section 7.2.

Section 2 formally introduces the group inefficiency ratio, and Section 3 compares it to existing definitions. Section 4 computes overall and group inefficiency for fourteen voting rules on synthetic elections with cohesive blocs, covering rules common in practice as well as rules developed for their mathematical properties. Section 5 makes the Borda/STV contrast precise with asymptotic results for well-separated blocs: STV achieves the best possible group inefficiency ratio of 1, while for Borda it diverges to infinity. Finally, Section 6 sweeps a large dataset of real elections, where the measurement differences are attenuated but the geometry observed in the synthetic experiments often carries over intact.

2 Preliminaries

2.1 Elections, profiles, and voting rules

Suppose we have a set $\mathcal{V}$ of n voters and a set $\mathcal{C}$ of m candidates. An *embedded election* is a metric space X with distance function d, equipped with an embedding map $\mathcal{M} : \mathcal{V} \cup \mathcal{C} \to X$ that identifies the voters and candidates with points in the space.[1] This gives us a way to measure how different two candidates are, and how close a voter is to a candidate, by identifying each v and c with its image in X. The associated *preference profile* $P = P(X, \mathcal{M})$

[1] This could also be called a *metrized* election; it just refers to a choice of metric space that holds images of the voters and candidates.

is a set of rankings $\succcurlyeq_v$ for each $v \in \mathcal{V}$, where for any voter v and pair of candidates c, c', we have $c \succcurlyeq_v c' \iff d(v,c) \leq d(v,c')$, meaning that voters prefer the candidates located closer to themselves. Importantly, the profile is strictly less information than the embedded election, in the sense that the profile can be uniquely derived from the embedding but not vice versa.

If $\mathcal{P}$ is the set of profiles of a given type, then $f : \mathcal{P} \to 2^{\mathcal{C}}$ denotes a multi-winner voting rule (otherwise known as an *electoral mechanism* or a *social choice function*). We will focus on rules designed to elect a fixed number k of winners, i.e., where $|f(P)| = k$.[2] Alongside deterministic rules of this kind, we will also consider rules that incorporate some randomization. The study of *metric distortion*, which has attracted great attention since its introduction circa 2015, evaluates the performance of a voting rule on an embedded election by considering the ratio of summed distances (or "costs," in economic terms) for its chosen winner(s) versus the alternative set that is optimal in X. The performance of a voting rule on a profile can then be measured by the worst case over all embeddings consistent with that profile. (Here we do not take that worst case; all of our ratios are computed in a fixed embedding.) The foundational work is surveyed in Section 3. Here, we build on the cost ratio from the distortion literature to assemble different notions of fairness, centering a group fairness metric that assesses outcomes against a standard of proportional representation.

We use these metrics to compare a suite of voting rules. Two important examples are *plurality block voting* (Block) and the *Borda count* (Borda), which are positional rules that let voters award equal support to their top k choices or linearly decreasing support from first place to last, respectively. Another focal example, important in practical settings but requiring some technical detail to fully describe, is *single transferable vote* (STV), which uses a quota such as $q = n/(k+1)$ votes (called the *Droop quota*) and proceeds by rounds to elect any candidate over quota or eliminate the least popular candidate if needed, transferring votes after each event, until k seats are filled. *Ranked pairs* uses margins of victory to elect candidates who win many head-to-head comparisons by high margins; *single non-transferable vote* (SNTV) tallies only first-place votes. The full sweep of examples includes five voting rules that were expressly designed to be proportional or are regarded as "semi-proportional" (STV, SNTV, Chamberlin–Courant, Monroe, and Expanding Approvals), three randomized rules that are proportional only in expectation (OMRD, SMRD, DMRD), and five rules with different design goals (Block and Ranked Pairs are majoritarian; Borda is a so-called "excellence" rule; a pair of Veto rules are optimized for different notions of cost). A Uniform Random rule provides a baseline. Full definitions and citations are given in Appendix A.

2.2 Blocs, costs, and candidate assignments

Elections are called *polarized* (in everyday English) when there are sufficiently large, disjoint groups of voters whose voting behavior and preferences are sharply different. In the metric setting, we can easily model this by dropping groups of voter points centered at different locations. We call subsets of voters $\mathcal{B} \subseteq \mathcal{V}$ voting *blocs*. For any given bloc $\mathcal{B}$ and set of candidates S, we then define the cost of S to $\mathcal{B}$ as the sum of distances between voters and candidates:

$$\text{cost}(\mathcal{B}, S) = \sum_{v \in \mathcal{B}} \sum_{c \in S} d(v,c).$$

[2] More precisely, we consider rules that are guaranteed to elect k winners when candidates and voters are in general position. This allows for rules for which ties are possible, as long as they resolve when the embedding is perturbed.

This is in keeping with the dominant choice of objective function in the single-winner metric-voting literature, where a voter's cost for a candidate is given by distance and the societal cost is the sum of the distances from all voters to each winner.

Next, we also consider assigning a collection of voters to a subset of preferred candidates who (in informal terms) could best represent them. Given a bloc $\mathcal{B} \subseteq \mathcal{V}$ and a candidate set $S \subseteq \mathcal{C}$, their *candidate assignment* is a function $\Phi : 2^{\mathcal{V}} \times 2^{\mathcal{C}} \to 2^{\mathcal{C}}$ satisfying $\Phi(\mathcal{B}, S) \subseteq S$. This is thought of as identifying a designated subset of candidates from S that is best representative of the voters in bloc $\mathcal{B}$.

The choice of assignment function Φ may be varied, but in this paper we focus on a greedy candidate assignment that we denote by $\Phi = \text{Rep}$, keyed to the intuition of proportional representation.[3] For voting rules electing k winners, we can set the representational entitlement of bloc $\mathcal{B}$ however we want, such as by an intuitive option $r(\mathcal{B}) = \text{round}\,(k|\mathcal{B}|/n)$ that rounds their raw proportional ratio $k|\mathcal{B}|/n$ to an integer. We will use nearest-integer rounding $\text{round}(x) = \lfloor x + 1/2 \rfloor$ as our default.[4]

Using this, we can assign to a bloc of voters the cost-minimizing set of candidates of that size. For $T \subseteq \mathcal{C}$ we define

$$\text{Rep}(\mathcal{B}, T) := \operatorname*{arg\,min}_{\substack{S \subseteq T \\ |S| = r(\mathcal{B})}} \text{cost}(\mathcal{B}, S).$$

For example, if a voter bloc makes up 20% of the electorate in a 4-winner election, then Rep picks out their single most preferred winner, i.e., the winning candidate who minimized the summed cost to that bloc. Blocs smaller than 12.5% of voters are assigned to the empty set in a 4-winner scenario.

2.3 Group-centered fairness

Recall that $\mathcal{C}, \mathcal{V}$ are the sets of candidates and voters respectively, and let $\mathcal{W} = f(X)$ be the winner set given an embedded election and a voting rule.

▶ **Definition 2.1.** *For a voting rule f, entitlement function r, candidate assignment function Φ, and voter bloc $\mathcal{B} \subseteq \mathcal{V}$ in an embedded election X, define the* group inefficiency ratio *by*

$$\mathcal{I}(\mathcal{B}) = \mathcal{I}_{f,r,\Phi,X}(\mathcal{B}) := \frac{\text{cost}(\mathcal{B}, \Phi(\mathcal{B}, \mathcal{W}))}{\text{cost}(\mathcal{B}, \Phi(\mathcal{B}, \mathcal{C}))}.$$

Below, we use nearest-integer r and cost-minimizing $\Phi = \text{Rep}$ unless stated otherwise. We say f has overall inefficiency ratio α *if $\mathcal{I}(\mathcal{V}) = \alpha$, which is the standard notion of cost ratio used to define metric distortion in computational social choice theory.*

In Figure 1 we display computations for a few commonly used multi-winner voting rules: Block, Borda, and STV.

While worst-bloc inefficiency is an appealingly neutral notion, the worst-off subset is often an arbitrary-looking slice of the electorate that is hard to describe, such as a combination of pieces of the natural groups, or just a neighborhood of a certain loser (Figure 1). The north and south clusters of voters—making up the full electorate—are each well served by

[3] "Greedy" here means choosing the candidates of lowest cost regardless of their location, separation, etc.

[4] We also conducted extensive experiments with the round-down convention $\text{round}(x) = \lfloor x \rfloor$, but these had undesirable properties. With two blocs it will generically yield an entitlement gap, while nearest-integer rounding makes entitlement generically sum to k when bloc sizes are close to $1/k$ increments.

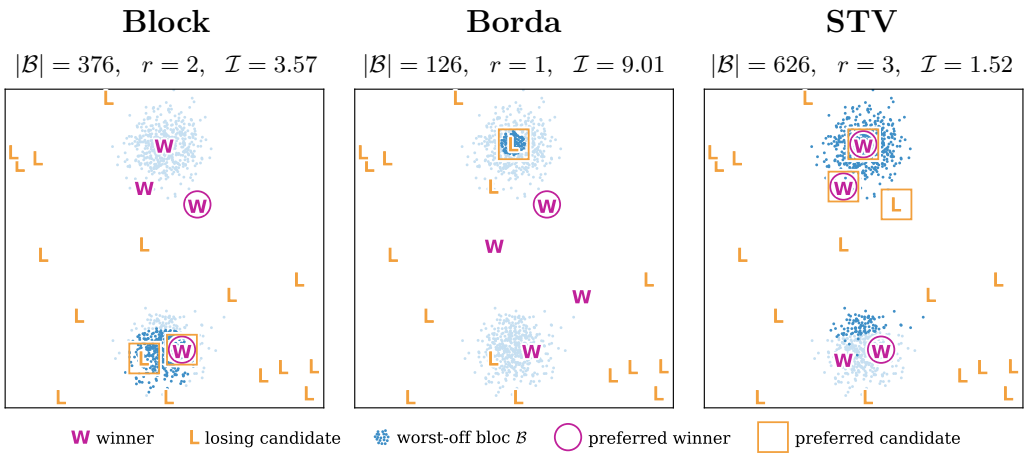


Figure 1 A single embedding of 1001 voters, with coinflip assignment to two voter blocs normally distributed around $(0, \pm 2)$ with $\sigma = 1/3$. Twenty candidates are drawn uniformly from the square. Four winners are computed by Block Plurality ("Block"), Borda, and STV, in turn, with W and L markers for winning and losing candidates. An exact optimization is run to find the worst-off bloc $\mathcal{B}$ of any size, shown in dark blue; the sizes proportionally entitle that bloc to $r = 2, 1$, and 3 winners, respectively. The group inefficiency ratio $\mathcal{I}$ for that worst-off bloc is shown each time.

their designated STV winners, but some combined groupings look worse-off. Dually, the two clusters are both poorly served in the Borda example but the worst-case analysis focuses on ardent supporters of a single candidate. Neutrality over all subsets should not be expected to find the externally obvious structure, and it does not. In Section 4 we will see that this also blurs systematic differences between rules, which appear clearly once the salient groups are fixed.

3 Related work

Our study branches off from a line of recent work that evaluates voting rules in metric settings. For single-winner voting rules, Anshelevich and Postl [3] and Anshelevich et al. [2] introduced the metric framework and the objective of seeking low-distortion voting rules, proving bounds for both known and novel voting rules. This opened a vibrant line of work studying metric distortion of voting rules, which in several cases led to the introduction of new rules that are of significant independent interest [10, 11, 25, 26]. Because the metric framework assigns numerical costs to outcomes, it allows finer-grained comparisons between voting rules than traditional axiomatic approaches.

The case of multi-winner elections is sometimes referred to as "committee voting," though we will avoid that terminology here because we are motivated by the case of political representation. Multi-winner voting rules have recently seen increased attention, with various authors studying their axiomatic properties [16, 17]. The study of proportionality axioms for multi-winner rules, in particular, is very active, and we refer readers to [6] for both a comprehensive review and recent developments. However, despite being interesting and applicable, multi-winner elections have received much less attention in the metric voting literature. Results from our work are strongly inspired by Elkind et al. [15], where the authors use simple metric embeddings in order to produce intuitive visualizations, giving qualitative, geometric insight into how multi-winner voting rules represent voters' preferences.

We build upon their work by defining fairness measurements and using them to evaluate the performance of several voting rules in various metric embeddings.

Prior work on multi-winner voting with similar notions of distortion, cost, and efficiency includes the following.

- **Proportional fairness in utility distortion.** Alongside optimal distortion bounds, Ebadian et al. [14] study a proportional-fairness requirement that every cohesive group of voters have influence on the outcome proportional to its size, and give a single-winner randomized rule achieving an optimal $\Theta(\log m)$ approximation to it.
- **Worst-bloc fairness.** Moving to the metric setting, Goel et al. [19] compare the cost to the worst-off b voters under the chosen outcome with the same quantity under the optimal outcome, and bound this ratio uniformly over all b. In their multi-winner extension, a voter's cost is the sum of distances to *all* winners. Every voter is therefore charged for every winner, so a winner set split between two clusters is costly to both; there is no notion of a bloc being entitled to only some of the seats, which is the proportional element in our definition.
- **Distortion with q-cost.** Caragiannis et al. [8] take a voter's cost to be the distance to their qth closest winner, summed over all voters. They find sharp phase transitions in q, with distortion unbounded for every rule when $q \leq k/3$. Small q rewards proportional outcomes, while large q rewards compromise: at $q = k$, homogeneous winner sets near the center of the whole electorate do well, as noted in [23]. Their work is a framework for distortion bounds rather than a comparison of how specific rules perform.
- **γ-proportional representation.** The recent paper of Kalayci et al. [23] is closest in spirit to our work. In their work (as in ours), sufficiently large blocs of voters are assigned a proportional share of representatives from the winner set, and cost ratios are computed. Their notion of γ-proportional representation strengthens the proportional fairness of Chen et al. [12] from clustering, and their main theorem shows that a version of Expanding Approvals achieves it at a constant level once blocs are slightly larger than their proportional threshold, i.e., with resource augmentation.

These definitions share a design choice: fairness is measured either by summing over the whole electorate or by taking the worst case over all voter subsets of a given size. Our group inefficiency ratio is set up to work with blocs picked out by an external criterion, and we show that this choice changes which rules look fair.

4 Synthetic experiments

4.1 Experimental design

Following Elkind et al. [15], we work in the Euclidean plane $\mathbb{R}^2$ with deliberately simple configurations: voters are drawn in distinct, cohesive blocs, so that there are obvious groups to measure from and the differences between rules are sharp. In each trial, we sample voter and candidate positions as specified below in Section 4.2, then rank candidates by distance to obtain each voter's ballot, run every rule on the resulting profile, and compute overall and group inefficiency for each winner set. Unless otherwise specified, each experiment uses 1000 trials, which is enough to get stable distributions.[5] Results for the principal experiment are shown in Figure 2.

[5] See `https://github.com/mggg/metric-voting` for all code used in this paper.

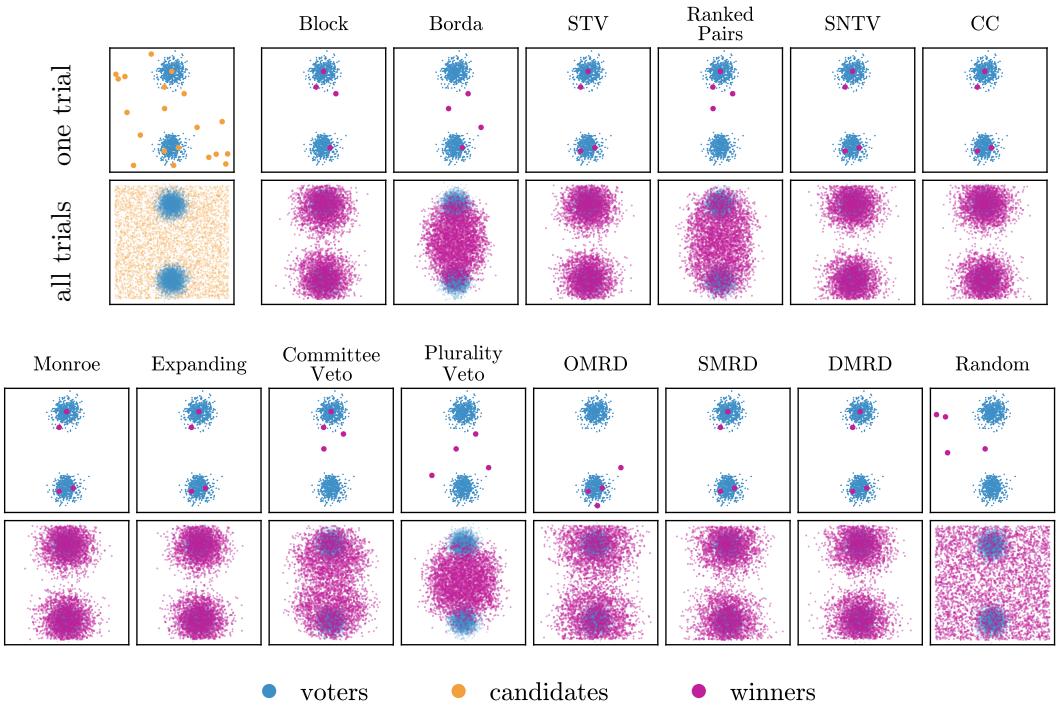


Figure 2 Echoing the visual in [15], this plot shows the trends in winner location for 13 multiwinner voting rules and a final alternative that chooses its four winners uniformly at random. For each voting rule, the upper plot shows the same contest from Figure 1; for the lower plot, we now pool 1000 independent trials and depict the 4000 winner locations together.

The main cleavage is clear, dividing the voting rules that favor compromise from those that instantiate power sharing. Pooled over many trials (Figure 2), there are visibly different families. Borda, Plurality Veto, and Ranked Pairs place a large share of their winners in the compromise zone between the blocs (62%, 71%, and 48% of winners within distance 1 of the horizontal midline, against 33% of candidates). The rules geared for proportionality do the opposite, placing fewer than 10% of winners there. Even the manifestly unreasonable OMRD, in which a single voter chooses the entire winner set, aligns winners to the bloc loci more often than chance.

Interesting differences are observable within the compromise family. In Figure 2, Plurality Veto concentrates its winners at the center, while Borda and especially Ranked Pairs spread them along the axis joining the blocs. The plots in Appendix B highlight further differences. With many voters and candidates (Figure 7), Borda winners collapse onto the midpoint, while Plurality Veto and Committee Veto form two lobes between the blocs, short of either bloc center. With four blocs and five seats (Figure 8), the compromise rules, now joined by Block, fill out an X shape with winners at the pairwise compromise points between adjacent voter clusters.

4.2 Experiment list

1. **Basic two-bloc experiment.** Two blocs, $n = 1001$ voters, $m = 20$ candidates, $k = 4$ winners. Bloc assignment by coinflip, with blocs normally distributed with means

$\mu_i = (0, \pm 2)$ and standard deviation $\sigma = 1/3$. Candidates are uniformly distributed on the 6×6 square centered at $(0, 0)$.

2. **Varying bloc size.** Voters and candidates are distributed as in Experiment 1, but voter bloc sizes are varied, starting with 0 voters in the first bloc and shifting 5% of the electorate at a time until all 1001 voters belong to the first bloc. Results for each setting are computed for 500 random trials.
3. **Four blocs, four winners.** Four blocs, $n = 1001$ voters, $m = 20$ candidates, $k = 4$ winners. Voters assigned to a bloc uniformly with centers $(0, \pm 2), (\pm 2, 0)$, with $\sigma = 1/3$.
4. **Many candidates, many voters.** Two blocs, $n = 10,001$ voters, $m = 200$ candidates, $k = 4$ winners, all else as in Experiment 1.
5. **Overlapping blocs.** Setup identical to Experiment 1 except that $\sigma = 1$ to make for more substantial overlap.
6. **Four blocs, five winners.** Setup identical to Experiment 3 except for choosing $k = 5$ winners while there are four blocs.

4.3 Findings

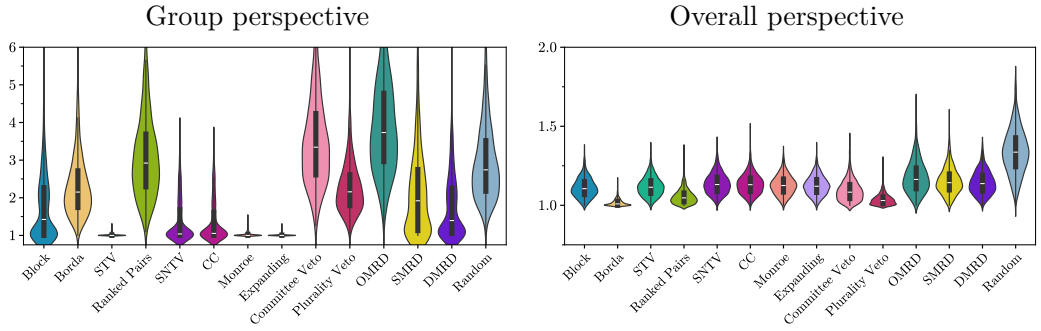


Figure 3 Experiment 1 (two blocs of equal size). Violin plots show the distribution over 1000 trials of group inefficiency $\mathcal{I}(\mathcal{B})$ and overall inefficiency $\mathcal{I}(\mathcal{V})$ for each rule. Here group inefficiency is $\mathcal{I}(\mathcal{B})$ for the worse-off of the two natural voter blocs, which divide the electorate into roughly equal and cohesive halves. Overall inefficiency ratios stay below 2 for every rule, while group inefficiency ratios range much higher and separate the rules sharply, giving a very different view of fairness. Some violins appear to extend below 1; this is an artifact of smoothing, where many values sit at 1 and the spread is wide. Inefficiency values are never smaller than 1.

- **The group perspective highlights proportional performance on a per-contest basis.** The distributions in Figure 2 show many rules looking essentially equally effective at providing winners centered on the north and south blocs—Block, STV, SNTV, CC, Monroe, Expanding Approvals, and even two Random Dictator variants. But this kind of view can't show whether each outcome is split proportionally across the blocs or whether the even appearance is due to aggregation across trials. The group inefficiency measurements in Figure 3 are needed for this. STV, Monroe, and Expanding Approvals have by far the strongest performance, with group inefficiency ratios near 1. The same three rules, however, appear ordinary among the larger set of voting rules from the whole-electorate, overall perspective.
- **The group perspective separates the proportional rules.** STV, Monroe, and Expanding Approvals split the seats 2–2 in over 99% of trials, compared to under 50% for

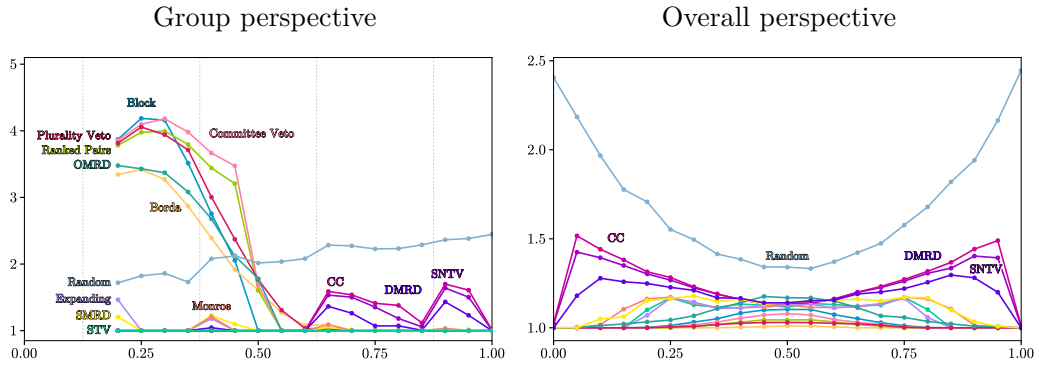


Figure 4 For Experiment 2 (two blocs of varying sizes), we plot the inefficiency ratio over different bloc sizes. We vary the point-of-view bloc in increments of 5%, performing 500 trials at each level and plotting the median $\mathcal{I}$ value. The thresholds where rounded entitlement transitions from $r = 0$ to $1, 2, 3, 4$ are shown. Overall inefficiency once again tells a strikingly different story from the one that emerges when we center fairness to a distinguished voter bloc.

Block voting. Chamberlin–Courant and SNTV fall in the middle, getting the 2–2 split about 70% of the time.

- **CC, DMRD, and SNTV stand out in both perspectives.** In the group view they spike when the bloc is a clear majority and in the overall view they spike near the extremes—both are cases where most rules are near 1. These spikes reflect the same tendency to give the smaller bloc an extra seat, because each rule has diminishing returns for a bloc's additional winners.[6]
- **The worst-case analysis loses the signal.** A definition based on the worst case over all subsets of voters, such as the proportional representation definition of [23], would hide the markedly strong group inefficiency performance of STV, Monroe, and Expanding Approvals in a polarized setting, since the whole-electorate inefficiency ratio is worse than for the natural choice of blocs, and $\mathcal{B} = \mathcal{V}$ is a candidate in the supremum.
- **Who bears the cost depends on the rule.** When one bloc has less than half of the electorate, Figure 4 shows that the five rules not designed for proportionality (Block voting, Plurality Veto, Committee Veto, Ranked Pairs, and Borda) have spiking group inefficiency ratios, along with OMRD among the randomized rules. When one group dominates in size, Block voting, for example, provides little check on their voting power—a cohesive group with even a slight majority can claim more than its share of seats, and the smaller group's inefficiency ratio jumps. CC, SNTV, and DMRD err in the other direction, actually under-representing the *larger* bloc just past the thresholds for earning three or four seats. STV is the only rule whose median stays at 1 across the whole plotted range. These patterns are not reflected by measurements from the overall perspective.
- **The pattern is robust and grows more pronounced at growing scale.** The same division between the perspectives appears in every experiment (Experiments 3–6, Appendices B–C): with four cohesive blocs, with overlapping blocs, and with an odd

[6] CC counts only each voter's closest winner, so a third seat near the majority gives less reward than a first seat for the minority (vs. Monroe, where balance constraints prevent this). DMRD discounts voters whose favorite has been elected, which happens more often in the larger bloc (vs. SMRD, which has no discount). SNTV splits the larger bloc's first-place votes among nearby candidates, so its third or fourth choice can trail the smaller bloc's second (vs. STV, which transfers the surplus).

number of seats, STV, Monroe, and Expanding Approvals perform strongly in group inefficiency ratios, while Borda comes with a steep cost. Letting the blocs overlap (Exp. 5) compresses the differences, but the ordering of rules is largely preserved. With more voters and candidates (Exp. 4), the gap between compromise and proportional rules widens. This will be examined asymptotically in the next section.

5 Theoretical results

Our next task is to provide abstract reasons for some of the differences in performance. We will focus on the two rules that most clearly exemplify the impact of the perspective shift between salient groups and overall voters: Borda count (best from overall perspective, followed by Plurality Veto) versus STV (best from group perspective, along with Monroe and Expanding Approvals).

Let $\mathcal{W}_S$ be the winner set under STV (with fractional Droop quota $q = n/(k+1)$), and $\mathcal{W}_B$ the winner set under Borda (with standard scores). For a multiset S in X, let $\min_{\ell,S}(x)$ be the ℓ points from S nearest to x. We will just write $\min_\ell(x)$ when S is understood to be the candidate set $\mathcal{C}$. Write $B_r(x)$ for the (open) ball of radius r based at x.

▶ **Theorem 5.1** (Convergence for point-mass voters). *Let $X = \mathbb{R}^2$ with the Euclidean metric $d(x, y)$ and fix distinct points b, h with midpoint $e = \frac{b+h}{2}$ at distance $\Delta = d(b, h)/2$ from each, and let Ω be a bounded, measurable domain containing $B_{3\Delta}(e)$. Suppose we independently draw n voters from $\mu_v = \frac{1}{2}\delta_b + \frac{1}{2}\delta_h$, and m candidates from $\mu_c = \mathrm{Unif}(\Omega)$. Fix $k = 4$ winners. Then for any $\epsilon > 0$, as $m, n \to \infty$,*

$$\mathbb{P}\left(\mathcal{W}_S = \min_2(b) \cup \min_2(h)\right) \to 1 \qquad \text{and} \qquad \mathbb{P}\left(\mathcal{W}_B \subset B_\epsilon(e)\right) \to 1.$$

In particular, the STV winners converge in probability to an even split between the voter locations b and h, while the Borda winners converge in probability to the midpoint e.

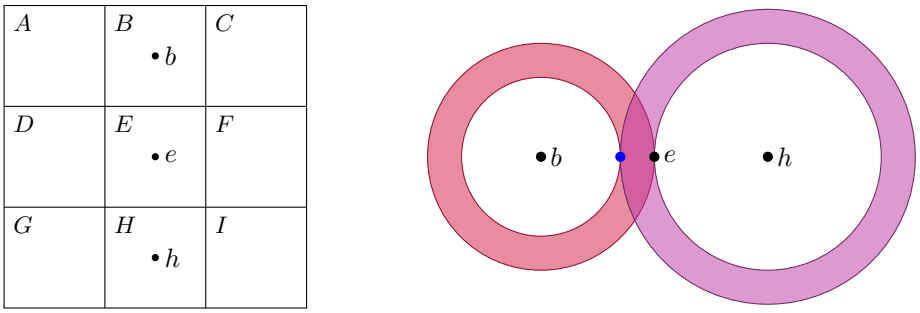


Figure 5 At left, a configuration of points that resembles the synthetic experiments. At right, an illustration of the optimality of e in terms of Borda scores: the blue dot shows an alternative position, which has a gain in Borda points (red area) outweighed by its loss (purple area).

Proof. Fix a small $\epsilon > 0$. Then the probability of a candidate landing in $B_\epsilon(b)$ is $\pi\epsilon^2/\lambda(\Omega) > 0$ (for the area measure λ), and therefore at least two candidates fall in that ball with high probability.[7] Taking $\epsilon < \Delta$ ensures that the two candidates nearest b and the two nearest h are distinct with high probability. The threshold for election is $q = n/5$, and the i.i.d.

[7] "With high probability" is used throughout to mean probability tending to 1 as $m, n \to \infty$.

draw means that the number of voters at b converges to $n/2 = \frac{5}{2}q$, so that there are at least $2q$ voters at b with high probability. Since all ballots of b voters are identical, the nearest candidate $\min_1(b)$ is over threshold, so they are immediately elected, transferring at least q weight to the next one and causing both of the two closest candidates to be elected. The same is true for h voters, which completes the proof for $\mathcal{W}_S$.

For the Borda statement, we will show that the midpoint e maximizes the expected Borda points. To see this, first consider the viewpoint of a voter located at b. Note that only candidates in $B_\Delta(b)$ are ranked higher by b than someone in position e. Thus the share of candidates ranked higher is roughly $\pi\Delta^2/\lambda(\Omega) < 1/2$, and likewise for a voter at h, so the expected e ranking is above average for both types of voter. Suppose some location receives a higher Borda score in expectation. Then it must be closer to one or the other of b, h than e is; without loss of generality, it is closer to b, say at distance $\Delta - \delta$ from b (see Figure 5). Of all such points, the one with the highest ranking from h is along the segment $\overline{bh}$. The expected gain in ranking from b is proportional to the area of the annulus of outer radius Δ and inner radius $\Delta - \delta$; on the other hand, the loss from the h voter goes with the annulus of outer radius $\Delta + \delta$ and inner radius Δ. This is a net loss in ranking, and therefore in Borda points, which shows that e is optimal. By the law of large numbers, the realized Borda scores are uniformly close to the expected scores, since both are determined by counts of candidates in balls around b and h. Since e is the unique maximizer of the expected score, with a definite gap outside $B_\epsilon(e)$, all four winners lie in $B_\epsilon(e)$ with high probability. ◀

▶ Remark 5.2. This easily generalizes to any $X = \mathbb{R}^d$ $(d \geq 2)$ and any even k, with only the volume calculations and the quota arithmetic getting messier. For odd $k = 2j + 1$, the same proof guarantees that $\min_j(b)$ and $\min_j(h)$ will be elected, with the final seat going to the $j + 1$st closest candidate for one of the blocs. The argument carries over from the fractional Droop quota to the integer quota used in the experiments, at the cost of slightly more notation. This proof strategy also works when the size requirement on Ω is relaxed, as in the experiments, but requires more calculation to handle boundary effects.

▶ **Corollary 5.3** (Group inefficiency under asymptotic conditions). *As $n, m \to \infty$, we have $\mathcal{I}_S(\mathcal{B}) = 1$ with high probability while $\mathcal{I}_B(\mathcal{B}) \to \infty$ in probability. This holds for either bloc $\mathcal{B}$ (the voters based at b or at h).*

The first statement holds because the group inefficiency is 1 when the preferred $r(\mathcal{B}) = 2$ candidates for the bloc are winners, as in STV. On the other hand, the cost ratio for Borda has a numerator of roughly 2Δ per voter, while the contribution to the denominator tends to zero.

Next, assume voters are drawn from normal distributions centered at b, h, as in the synthetic experiments.

▶ **Theorem 5.4** (Convergence for Gaussian voters). *Take X, Ω, μ_c, k as above. Suppose each voter i independently receives a bloc label $Z_i \in \{b, h\}$, uniformly at random, and a position $V_i \sim \mathcal{N}(Z_i, \sigma^2 I)$. Then as $m, n \to \infty$, we again have the conclusions of the point-mass case as the Gaussians concentrate: if $\sigma = o(m^{-1/2})$, then $\mathcal{W}_S = \min_2(b) \cup \min_2(h)$ with high probability, and $\mathcal{W}_B$ converges to e in probability.*

Proof. Fix $\epsilon = \epsilon(m)$ such that $\sigma \ll \epsilon \ll 1/\sqrt{m}$. Conditional on the label $Z = b$, the fraction of voters outside $B_\epsilon(b)$ goes to zero by the law of large numbers. Since the expected voter weight with $Z = b$ is $n/2 = \frac{5}{2}q$, the voter weight in the ball exceeds $2q$ with high probability.

Consider the gaps $d(b, C_{i_1}) < d(b, C_{i_2}) < d(b, C_{i_3}) < \dots$ from b to the nearest candidates. The ℓth gap is of order $1/\sqrt{\ell m}$. Choosing $\epsilon \ll 1/\sqrt{m}$ ensures that the difference between

$d(b, C_{i_2})$ and $d(b, C_{i_3})$ is greater than 2ϵ with high probability. By the triangle inequality, this means that the voters near b rank $\min_2(b)$ first and second, in some order. This, plus the fact that there are at least $2q$ such voters, is enough to ensure that they are elected before anyone can be eliminated.

For the Borda statement, it is enough to have $\sigma \to 0$ at any rate, so that expected Borda score converges uniformly to the expectation in the point-mass case. Since the point-mass case has its unique optimum at e, the optima in the Gaussian case converge to e, and so do the Borda winners. ◀

6 Observed data: Scottish elections

We complement the synthetic experiments with a large-scale analysis of Scottish local government elections, based on a publicly available dataset of over 1000 records of STV contests from 2007–2022, with nearly every contest electing 3 or 4 winners.[8] We focus on the 717 contests with 7–14 candidates.

To make a sweep feasible, we take advantage of two recently available tools. First, we use two embeddings whose properties were studied in recent work of Duchin and Tapp [13]. One uses Borda points to get an embedding $\mathfrak{b}$ of an m-candidate contest into $\mathbb{R}^m$, while the other uses head-to-head comparisons to get an embedding $\mathfrak{h}$ into $\mathbb{R}^{\binom{m}{2}}$ similarly.[9] As those authors discuss, the L^1 (Manhattan) distance can then be used on each of those coordinate spaces, recovering the Spearman footrule and Kendall tau distances on complete ballots and extending naturally to partial ballots. We will use those to get embeddings of both $\mathcal{V}$ (via their ballots) and $\mathcal{C}$ (represented as bullet votes for themselves, i.e., ranking only one candidate and omitting all others).

Then, Duchin et al. have computed globally optimal 2-clusterings of the votes for both $\mathfrak{b}$ and $\mathfrak{h}$ in all 1070 elections and made them publicly available: centers are the pair of points that minimizes total L^1 cost, then the ballots are labeled by whichever center is closer.[10] This gives us two blocs per election; we can then compute inefficiency ratios for the winner sets produced by the voting rules. It might seem that Borda count has a structural advantage in the Borda embedding, and this turns out to be the case in both.

▶ **Lemma 6.1.** *The Borda voting rule has overall inefficiency ratio identically 1 in both embeddings.*

Proof. *Borda embedding.* Since $0 \le \mathfrak{b}(v)_c \le m-1$ for all c, and $\mathfrak{b}(c)$ has an $m-1$ in the c coordinate and zeroes elsewhere,

$$|\mathfrak{b}(v) - \mathfrak{b}(c)|_1 = \sum_{c' \neq c} \mathfrak{b}(v)_{c'} + \big(m - 1 - \mathfrak{b}(v)_c\big) = |\mathfrak{b}(v)|_1 + (m-1) - 2\,\mathfrak{b}(v)_c.$$

If we sum this over all voters for fixed c, we get terms that don't depend on c minus twice the Borda score of c. This is minimized for candidates with the highest Borda scores, i.e., the Borda winners.

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[8] See `https://github.com/mggg/scot-elex`. 554 contests were for 3 winners, and 508 for 4 winners, out of 1070 in all.

[9] For a ballot v that ranks some of the m candidates, the Borda embedding $\mathfrak{b}(v) \in \mathbb{R}^m$ fills in the coordinate i with the value $m - p$ if candidate i is ranked in position p. Unranked candidates score 0 (the "pessimistic" convention). The head-to-head embedding $\mathfrak{h}(v) \in \mathbb{R}^{\binom{m}{2}}$ has coordinates indexed by pairs of candidates, with values given by $\mathfrak{h}(v)_{cc'} = 1$ if $c \succ_v c'$, -1 if $c' \succ_v c$, and 0 if neither is ranked, where a ranked candidate is preferred to every unranked one.

[10] `https://github.com/mggg/scottish-ballot-clustering-replication/` (paper in preparation)

Head-to-head embedding. First, note $\mathfrak{h}(c)$ has a ± 1 in the coordinates involving c with the sign favoring c, and 0 in the other coordinates. Comparing with voter v's ballot, the L^1 difference is equal to $|\mathfrak{h}(v)|$ for coordinates not involving c, so it is 1 if v compared the two candidates and 0 if not. For coordinates involving c, the difference is 0 if $c \succ_v c'$, 2 if $c' \succ_v c$, and 1 if both are unranked. This means

$$|\mathfrak{h}(v) - \mathfrak{h}(c)|_1 = |\mathfrak{h}(v)|_1 + (m-1) - 2\,\#\{c' : c \succ_v c'\}.$$

But the count of candidates ranked below c is exactly the Borda score v awards to c. Summing over voters once again, we get that the cost minimizers are the Borda winners. ◀

Given these tools, we can compute $\mathcal{I}(\mathcal{B}_1)$, $\mathcal{I}(\mathcal{B}_2)$, and $\mathcal{I}(\mathcal{V})$ across the full 717-election dataset for all of our voting rules.[11] We should expect the differences in group inefficiency ratios to be greatly reduced compared to the stylized synthetic examples in which the groups were cohesive and well-separated; in real elections, they will be messier. We focus on the head-to-head embedding $\mathfrak{h}$ (for both the costs and the clustering) to reduce the structural advantage for Borda count that is "baked in" because $\mathfrak{b}$ is built directly from Borda scores—though by Lemma 6.1, Borda is optimal for the whole electorate in either embedding.

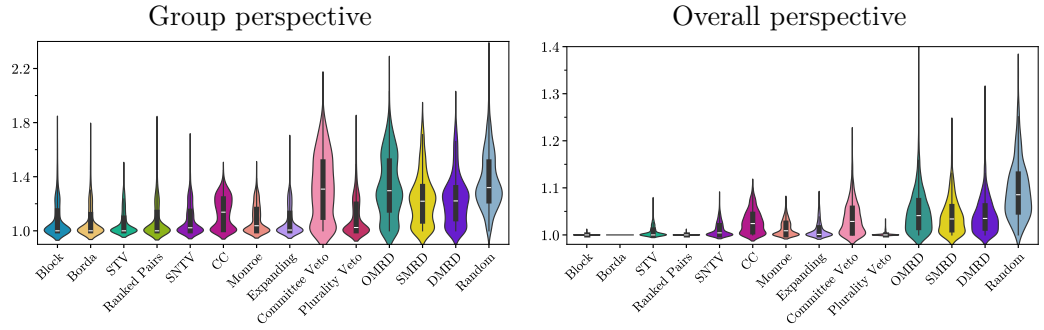


Figure 6 Of the 717 Scottish STV contests with at least seven candidates, we find the winners by all the voting rules studied here and compare the overall and group inefficiency ratios.

As expected, the differences between rules that were measurable in the stylized synthetic experiments are attenuated in real-world elections; nevertheless, Figure 6 shows that STV has the lightest upper tail of group inefficiency ratios among the rules we compared, with Monroe and Expanding Approvals also performing strongly. Taking the worse-off of the two natural blocs, the 95th-percentile inefficiency ratio is approximately 1.258 for STV, followed by Monroe and Expanding Approvals (both roughly 1.277), SNTV (1.296), Borda (1.306), and CC (1.314). The fact that even uniform random election has a 95th percentile figure of 1.737 makes it clear that differences will be simply smaller than in our earlier experiments, and should be interpreted accordingly.[12]

[11] Several rules need a convention for partial ballots; we treat unranked candidates as tied at the bottom throughout, with details in Appendix E.

[12] One factor contributing to more compressed score ranges is fewer candidates ($7 \leq m \leq 14$ instead of $m = 20$ in Experiment 1), which makes it less likely to have a candidate very close by. But much of this is due to the embedding: candidates are represented as bullet votes for themselves, so any voter ranking many candidates can't have a candidate too near, because truncation/extension contributes to ballot

Turning to Borda and STV—the comparison pair for the asymptotic theorems—one reason for small differences is that they elect precisely the same winner set in 387 of the 717 elections. If we further narrow the focus to the 330 contests in which their winner sets differ (shown in appendix Figure 10), the corresponding 95th percentiles are 1.259 for STV and 1.376 for Borda: the advantage for STV has widened.

The differences point the same way as in the synthetic experiments, with Borda excelling from the overall perspective. We have seen that it's provably optimal at $\mathcal{I} \equiv 1$, with Block Plurality, Ranked Pairs, and Plurality Veto not far behind.

Though differences in the group perspective are on a smaller scale, the key geometry from the synthetic experiments carries over. Within the 330 contests with a different Borda and STV winner set, 178 have a bloc split with $2 + 2$ entitlement, and of those, 171 have a single candidate differing, so that $\mathcal{W}_B = \{C_1, C_2, C_3, B\}$ and $\mathcal{W}_S = \{C_1, C_2, C_3, S\}$.

- S has the most first-place support among all candidates in $\mathcal{C}^* = \mathcal{C} \setminus \{C_1, C_2, C_3\}$ in 147 out of 171.
- Relative to the two cluster centers z, z' and a third point x, let the *triangle defect* be $d(x, z) + d(x, z') - d(z, z')$, i.e., the amount by which the triangle inequality differs from equality. This measures how much of a detour it is to visit x between z and z'. B has the smallest possible triangle defect in 121 out of 171 cases, usually the uniquely smallest of the candidates in $\mathcal{C}^*$. In other words, it tends to lie roughly *between* the centers of the blocs, as in the synthetic experiments.[13]

A candidate drawn at random from $\mathcal{C}^*$ would meet either criterion in about 33 of the 171 contests, so the 147 and 121 counts with these properties are meaningful.

These observations show that STV favors candidates with high first-place support: there are many voters for whom these are the most preferred, so the winners are near the locations that concentrate many voters. The alternative winner favored by Borda tends to lie between the bloc centers. This is the same contrast as we saw in the synthetic and the asymptotic results, now in real elections.

7 Discussion

7.1 Summary and implications

Our results clarify the fairness claims of proportional voting rules, distinguishing between them and quantifying how they stand apart from rules with other design goals. Rules such as Borda find compromise among the entire voter population, electing centrally located, homogeneous sets of winners. STV, Monroe, and Expanding Approvals, on the other hand, do well at electing diverse committees that secure representation bloc-by-bloc. On the Scottish ballots, the differences between rules are greatly reduced, but the top-line finding is stable: STV, Monroe, and Expanding Approvals have the lowest group inefficiency—lower than their proportionally-oriented cousins CC and SNTV—while Borda, the majoritarian rules, and Plurality Veto are stars in overall efficiency.

It is striking that STV and Expanding Approvals, both using a round-by-round structure that incorporates ballot data in a straightforward way, are comparable and often better

distance. This means that in a ten-candidate election, a bullet voter has an L^1 distance of 0 to their chosen candidate and 18 to all others, while a complete-ballot voter has a distance of 36 to their first choice ranging up to distance 54 to their last choice.

[13] Note that since $\mathfrak{h}$ is tied to Kendall tau, these cluster centers agree with what are called Kemeny 2-centers (minimizers for summed swap distance) in COMSOC.

in group inefficiency than Monroe, which is a global (and NP-hard) optimization of a representation cost. Of those two, STV is arguably more transparent in terms of how your preferences are used (your support sits with your first choice until they are elected or eliminated, then moves down your ballot), and it runs in linear time, while the reweighting step in Expanding Approvals adds a factor of k.

We also find that rules that stand out for strong performance in some kinds of election contexts are measurably unfair in others, suggesting that the choice of a fair voting rule may be situational and even fluid. Even so, the widely used Block Plurality voting rule looks bad across the board, and fares worst in the presence of a small but cohesive minority.

This analysis has real-world implications. Though OMRD is offered as an intentionally unreasonable system and Borda voting is rarely used for political representation, Block voting is one of the most frequently used systems for local election in the United States, such as for city councils and county commissions.[14] STV is currently used, or is being considered for adoption, in many parts of the world. In the United States, reformers often claim that STV will provide stronger proportional representation for minority groups than legacy systems such as first-past-the-post and Block voting.[15] This paper gives supporting evidence for this assertion from the perspective of distinctive voter blocs, both in synthetic elections and against naturalistically observed Scottish preference data.

7.2 Identifying salient groups and other future directions

The group inefficiency of a voting rule depends on which groups we designate. In Section 6 we took the blocs to be the two distance-optimized clusters of voters in the head-to-head embedding, but this is only one of many possible choices, and which groups should count is a central question raised by this work. Possibilities range from exogenous (defined by labels such as race or party registration or by membership in a group impacted by proposed legislation) to endogenous (defined by the ballots alone).

Many possibilities have at least some endogenous component. Real-world voting often involves measurable polarization; for instance, in the decades that followed the passage of the Voting Rights Act of 1965 (VRA) in the United States, a lineage of statistical techniques has been created and refined for quantifying polarization along racial and ethnic lines. Minority groups who have distinct candidate preferences have been entitled to protection under the VRA only if they are sufficiently cohesive (in the sense of being likely to support a common set of candidates over the course of many votes) and the majority bloc is also cohesive in a manner that blocks minority electoral opportunity.[16] Thus relevant groups are defined by both a race label and their voting behavior: a mix of exogenous and endogenous factors.

For purely endogenous identification, we can look to the ballots themselves, letting blocs emerge from voting behavior. Clustering, as in Section 6, does this, but at a risk of circularity: voters form a bloc because they are close together in the same metric that is then used to measure how well they are represented. Future work should consider more methods of finding

[14] Block voting is used for at least some city council seats in major cities like Cincinnati (all nine seats), Boston, Detroit, and Charlotte, among numerous other top-100 cities like Aurora, CO, Lincoln, NE, Fort Wayne, IN, and Chesapeake, VA. Boston has initiated a process to transition to IRV/STV, and Denver voted to transition away from Block voting in 2025.

[15] This was a claim made by reform advocates in Portland, Oregon, which recently moved to STV for the election of its city council [30]. For more general examples, see [1, 5, 21, 18].

[16] As we write in 2026, the VRA is still standing law, but sits precariously. The Supreme Court found for Black plaintiffs in Alabama in *Allen v. Milligan* (2023) [34], but more recent decisions in *Alexander* [32] and especially *Callais* [33] have made it harder and harder for plaintiffs to prevail.

blocs from ballot data. Ideally the measurements that identify the blocs might be in some sense formally independent of the measurement of their representational success.

One promising approach is to look for groups with *simple descriptions*. A few pairwise comparisons, such as preferring both of candidates A, B to both of C, D, may be enough to pick out important subgroups of voters, while representation is still measured over full ballots. This is a special case of identifying groups by a simple class of functions, such as small decision trees, as in the multicalibration framework of [20], where such groups are called "computationally identifiable." Classical proportionality axioms such as PSC (*proportionality for solid coalitions*) already work this way: a solid coalition is a group with a simple description (all voters ranking some set $S \subset \mathcal{C}$ above the rest), and the axiom requires that candidates from S be elected in numbers corresponding to the size of their solid bloc of supporters [4]. A metric analog would ask for rules whose group inefficiency is small for every group in a richer class of simple descriptions.

Beyond the choice of groups, the benchmark itself has limits: because $\mathcal{I}(\mathcal{B})$ compares against the best available candidates, it measures what a rule does with the candidates on offer, and a group with no nearby candidates can score well while being poorly represented. Candidate supply is a separate, and important, dimension of fairness. On the theory side, Section 5 explains the performance of STV and Borda with asymptotic results. Extending these results to general voter distributions, broader bloc structure, and more voting rules, or obtaining worst-embedding bounds in the spirit of Kalayci et al. [23], is a challenging research direction that would continue to convert folklore or anecdotal differences into theorems. Representative democracy has high stakes, and electoral reform has significant momentum, making this a rare opportunity for theorists to meet practitioner demand head-on.

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A Voting rules

A full suite of implemented voting rules is publicly available in GitHub.[17] The rules are presented here in the order that they appear in the figures. We remind readers that the rules may be sorted into three groups: proportional/semi-proportional (STV, SNTV, CC, Monroe, and Expanding Approvals), proportional in expectation (OMRD, SMRD, and DMRD), and rules with other design goals (Block, Ranked Pairs, Borda, Committee Veto, and Plurality Veto).

- **Block voting (Block):** Among m candidates, a ranked ballot contributes $+1$ to the score of each candidate ranked in the top k positions; we choose the k candidates with the highest scores. (This is sometimes called plurality bloc voting or plurality block voting.) This is a positional voting rule with the scoring vector $(1, \ldots, 1, 0, \ldots, 0)$.
- **Borda count (Borda):** Among m candidates, we default to a scoring where a ranked ballot contributes $m - j$ to the score of the candidate ranked in position $j \in \{1, ..., m\}$; the k candidates with the highest scores are elected. This is a positional voting rule with the scoring vector $(m-1, \ldots, 2, 1, 0)$. Where data includes partial ballots, unranked candidates receive zero points.
- **Single transferable vote (STV):** This is a multi-round process that uses a threshold or quota for election, often the Droop quota $q = \lfloor \frac{n}{k+1} \rfloor + 1$ used in the experiments here, to choose k winners from n ranked ballots.[18] Candidates are initialized with a vote weight equal to their number of first-place votes. If any candidate has support $w \geq q$, they are elected, and support transfers to the next choice on their voters' ballots with fractional weight $(w - q)/w$. If in a given round no candidate meets the quota, then the one with the least current first-place support is eliminated and the ballots headed by those candidates are transferred (retaining their full weight) to their voters' next choice. This continues until k candidates are elected.
- **Ranked pairs:** This rule can be efficiently defined (and computed) in terms of the *tournament graph*, a weighted directed graph on the candidates. Let w_{ij} be the number of voters that rank C_i above C_j (or rank C_i and omit C_j). Then if $w_{ij} > w_{ji}$, we draw an edge from C_i to C_j of weight $w_{ij} - w_{ji} > 0$. Begin with all edges deactivated, and turn them on in order of the largest margins (edge weights), rejecting only the edges that would form a directed cycle with active edges. There are one or more rankings compatible with the active edges at the end of this process, in the sense that $A \succ B$ only if there is no directed path $B \to A$. Using a tiebreaker if needed, one of these rankings is selected; the k ranked pairs winners are the first k candidates in that order. In the terminology of social choice, this rule has a Condorcet consistency property: if some k candidates beat all other candidates head-to-head, then they are the winners.
- **Single non-transferable vote (SNTV):** Each candidate receives a score equal to their number of first-place votes. The k winners are the k candidates with the highest scores. This is the $\ell = 1$ form of *limited voting*, in which voters may choose a number of candidates ℓ that is smaller than the number of seats. This is a positional voting rule with the scoring vector $(1, 0, \ldots, 0)$.
- **Chamberlin–Courant:** This rule elects the set of k candidates maximizing the sum, over voters, of the Borda score each voter gives to the closest candidate in the set [9]. A

[17] `https://github.com/mggg/metric-voting`

[18] Basing the threshold on $n/(k+1)$ is called the Droop quota, while replacing that with n/k would give the Hare quota.

voter is thought of as being assigned to (and represented by) their highest-scoring winner. Finding the CC winner set is known to be NP-hard [29]. We compute CC winners exactly by enumerating all k-committees.

- **Monroe:** This rule is nearly identical to Chamberlin–Courant, but with the additional constraint that each winner must represent disjoint sets of $\frac{n}{k}$ voters, rounded up or down to account for fractions [27]. This is again NP-hard. We compute Monroe exactly: for each committee, the optimal balanced assignment is a transportation problem, solved as an LP.
- **Expanding approvals:** Like STV, this is a multi-round process with uses a quota for election. Here we use the default quota from Aziz-Lee: $q = \frac{n}{k+1} + \frac{1}{m+1}\left(\lfloor \frac{n}{k+1} \rfloor + 1 - \frac{n}{k+1}\right)$, which lies just above the Droop quota [4]. All voters are initialized with weight 1. In round j, each voter approves their top j candidates. If any unelected candidate has support $w \geq q$, the one with the most support is elected, and each supporting ballot's weight is multiplied by $(w-q)/w$, so that exactly q weight is used up. If no candidate meets the quota, then j is increased by one. This continues until k candidates are elected. Kalayci et al. analyze a special case of this rule with quota $\lceil \frac{n}{k} \rceil$ in which supporters are removed outright [23].
- **Plurality veto** was recently introduced by [26] as a single-winner rule achieving best-possible metric distortion among deterministic rules. Candidates start with initial scores equal to their number of first-place votes. Then voters are arbitrarily ordered (whether randomly or deterministically) and each voter in turn is queried for their lowest-ranked alternative among those candidates with positive scores; that least-favorite candidate loses a point. The winner is the last candidate with a positive score. We extend this rule to the multi-winner setting by initializing with k-approval scores (as is done for the Block rule), and then sequentially iterating through the voter ordering until exactly k candidates remain.
- **Committee veto** is a multi-winner variant of Plurality veto proposed by [26], in which voters rank candidate sets of order k (i.e., potential winner sets). For each voter, the top k candidates in their ranking are taken to be a single candidate committee. These n k-element subsets are the 'candidates' in a single-winner Plurality Veto election. To create a preference profile, each voter ranks candidate committees according to their distance to the qth closest committee member. For our experiments, we default to $q = k$.

Finally, we consider three variants of the Random Dictator rule for single-winner elections, in which the winner is the first choice of a uniform random voter.

- **One-Shot Multi Random Dictator (OMRD):** A single voter is randomly chosen and their top k preferences are immediately elected.
- **Sequential Multi Random Dictator (SMRD):** Randomly select k voters, in order, to act as dictators. Sequentially elect their favorite candidate who has not yet been elected.
- **Discounted Multi Random Dictator (DMRD):** In every round, a voter is randomly chosen and elects their top (not-yet-elected) preference. Then all voters who would have selected the same candidate in that round have their voting power discounted by a multiplicative fraction $\rho \leq 1$, and the voter distribution is re-normalized for the next round. Note that a voter may be discounted multiple times. In the synthetic experiments, we execute DMRD with $\rho = 1/2$.

B Visualizing Experiments 3-6

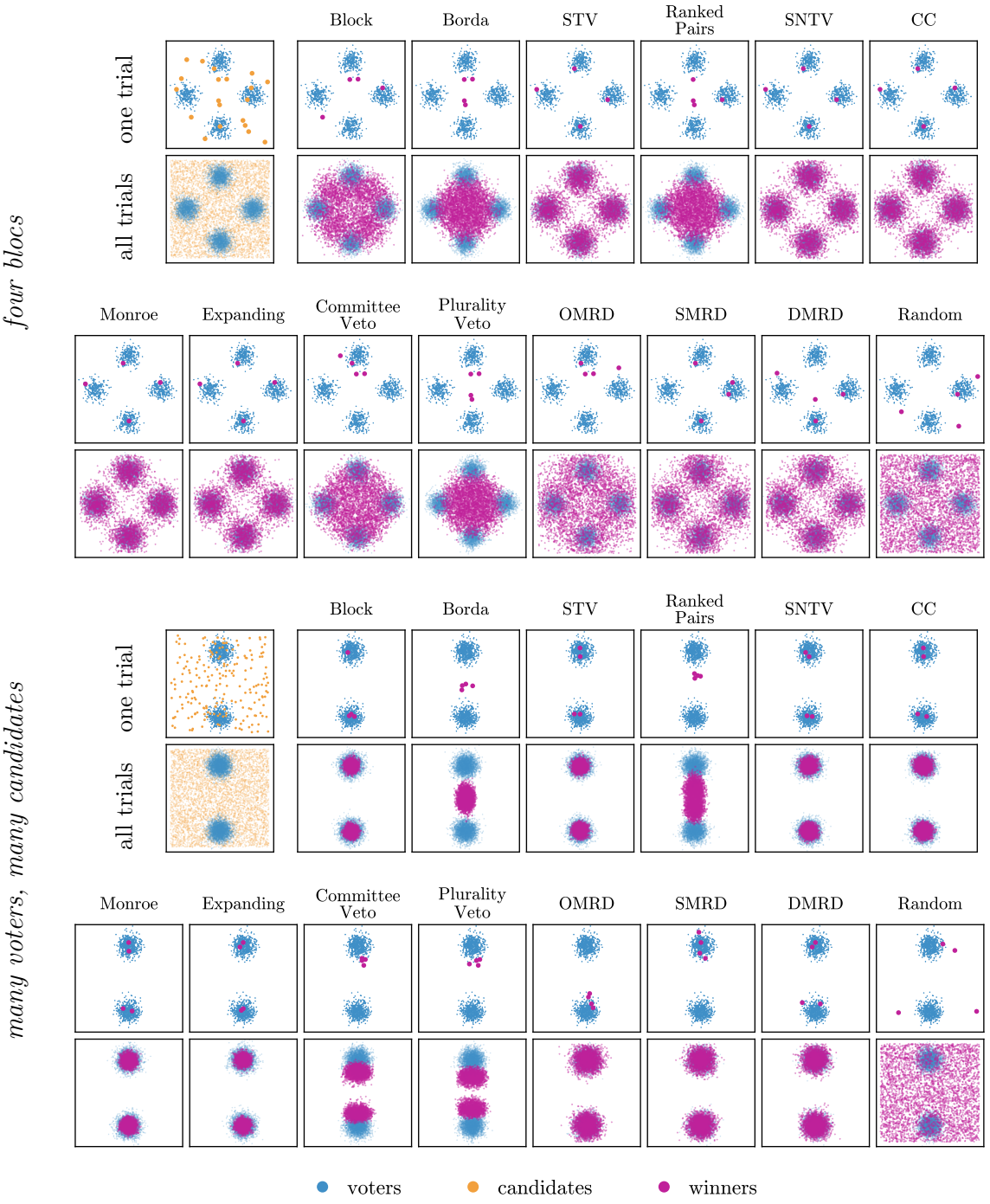


Figure 7 Single trials and pooled winner sets for Experiments 3-4 across 1000 trials.

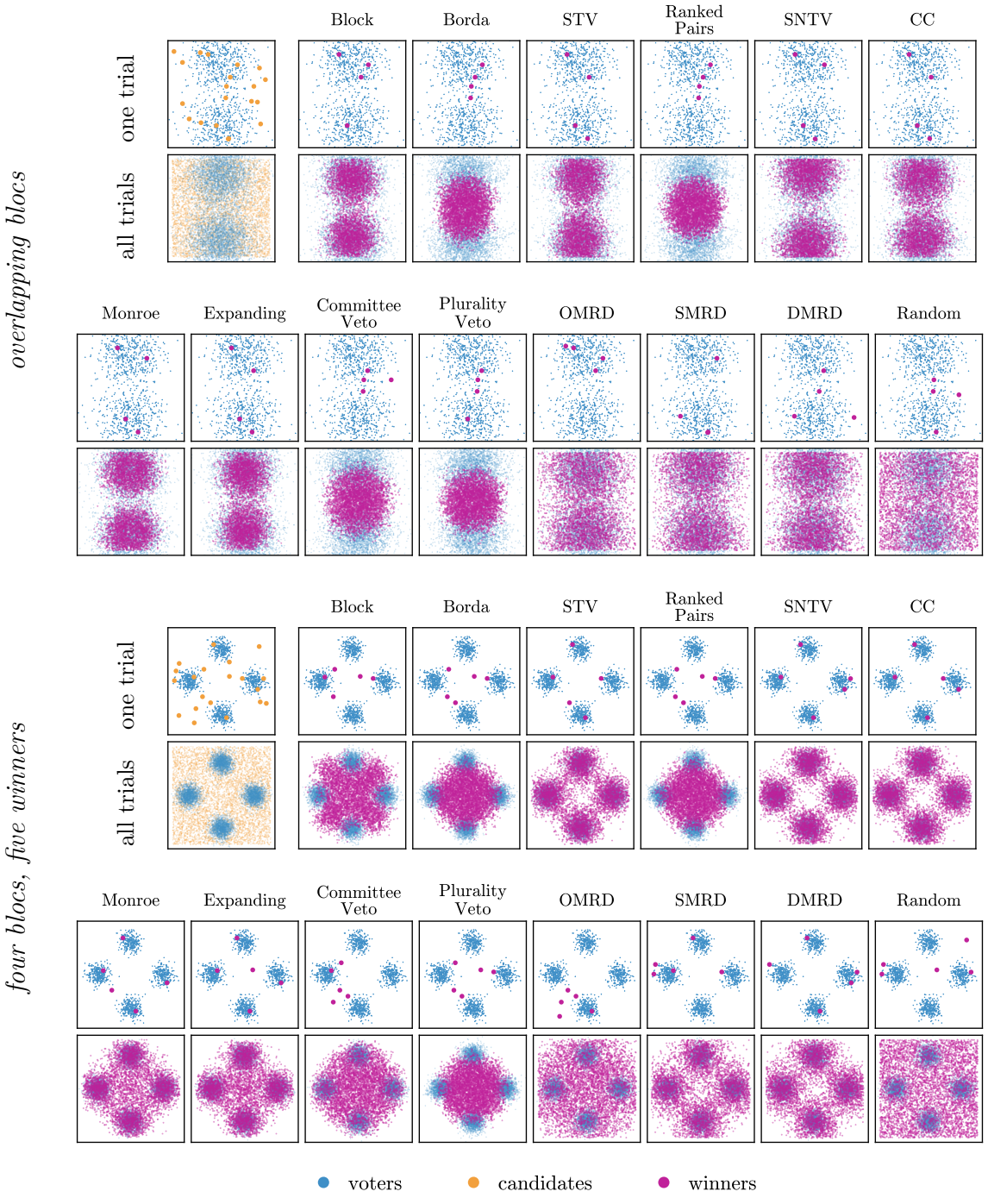


Figure 8 Single trials and pooled winner sets for Experiments 5-6 across 1000 trials.

C Summarizing group vs. overall perspective

For Experiments 4-6, we can now juxtapose the summary plots to emphasize the trend: the overall perspective consistently hides the enormously varying information from the point of view of the blocs from which the elections were generated. Violin plots show the distribution of values found in our random trials for group inefficiency ratios (left) compared to overall inefficiency ratios (right). In all of these, violin plots may create the appearance of values below 1.

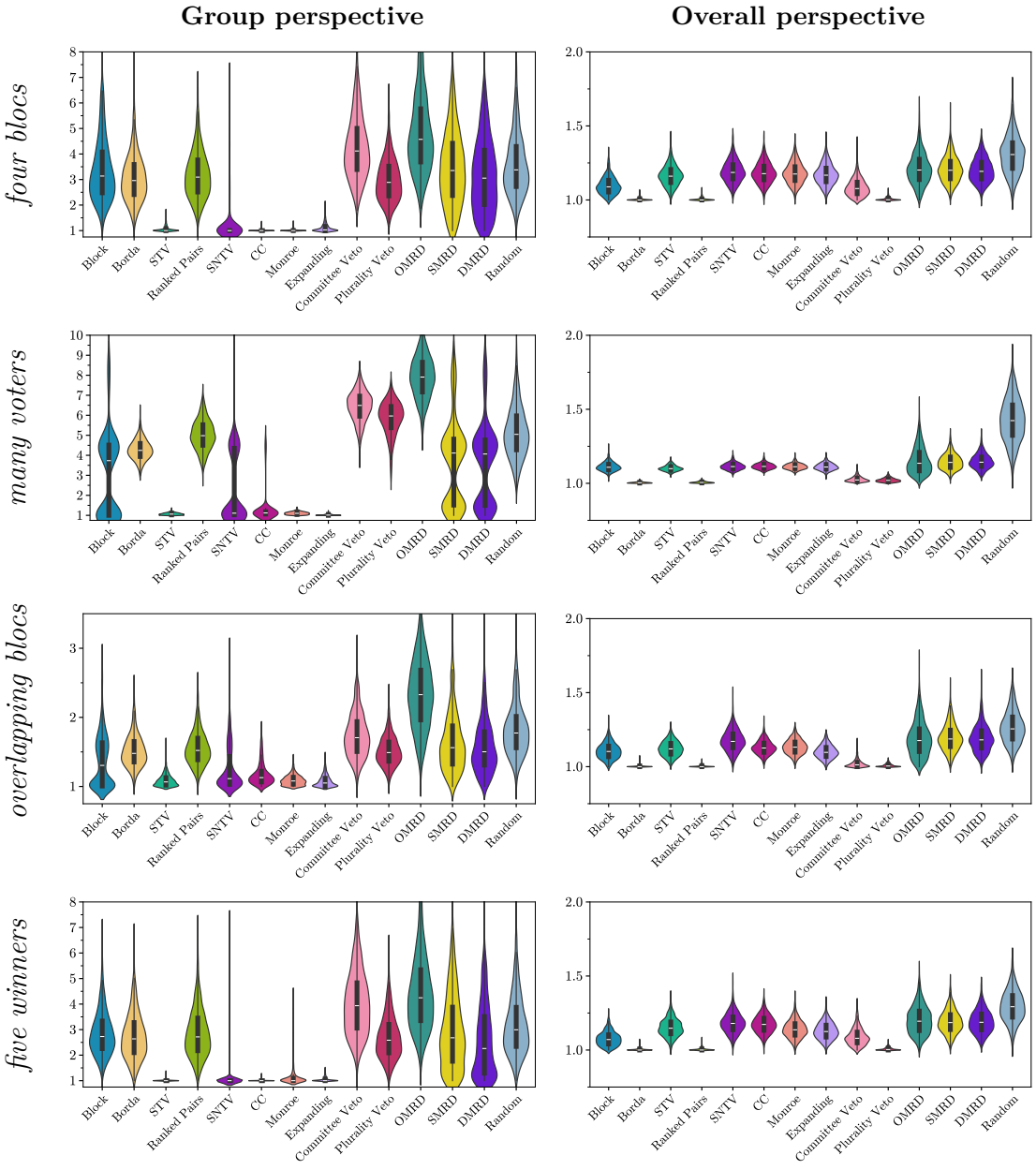


Figure 9 Experiments 3-6.

D Zone splits and more details on synthetic runs

																	
Block	**165**	98	138	38	50	24	129	70	17	12	4	·	20	14	6	16	*801*
Borda	3	·	34	1	15	45	2	2	·	87	**181**	82	35	8	2	·	*497*
STV	**367**	216	146	63	35	10	·	·	·	9	1	·	·	·	·	·	*847*
RkPairs	·	·	2	·	3	1	·	·	·	**118**	65	16	89	78	23	32	*427*
SNTV	**269**	156	96	51	36	5	159	47	6	2	·	·	2	1	·	·	*830*
CC	**269**	147	106	49	45	3	148	46	3	2	·	·	2	1	·	·	*821*
Monroe	**364**	227	135	65	33	12	·	·	·	5	1	·	·	·	·	·	*842*
Expanding	**382**	218	133	61	27	12	·	·	·	8	·	·	·	·	·	·	*841*
CmteVeto	·	·	·	·	·	·	·	·	·	73	23	5	77	**123**	74	62	*437*
PlurVeto	1	1	11	1	9	8	·	·	1	37	111	90	4	4	·	·	*278*
OMRD	·	·	·	·	·	·	·	·	·	35	10	·	53	110	**130**	126	*464*
SMRD	**113**	98	85	19	61	9	112	81	20	8	1	·	18	18	17	19	*679*
DMRD	**146**	132	115	33	59	10	123	85	14	5	1	·	6	4	5	5	*743*
Random	1	11	3	8	16	4	4	5	8	2	2	·	·	11	3	1	*79*

Table 1 For Experiment 1, this table shows the frequency (out of 1000 trials) for each of 16 winner configurations. Each icon shows the 3×3 zone grid of Figure 5; darker shading means more of the four winners in that zone. Configurations are identified with their images under both axial reflections. Boldface indicates the most frequent configuration for each rule, when it is among those depicted here. The last column gives the number of trials covered by the 16 configurations shown.

																	
Block	454	·	·	·	·	·	**473**	·	·	·	·	·	·	·	·	73	*1000*
Borda	·	·	·	·	·	·	·	·	·	5	11	**984**	·	·	·	·	*1000*
STV	**1000**	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	*1000*
RkPairs	·	·	·	·	·	·	·	·	·	104	224	**622**	42	·	·	8	*1000*
SNTV	**629**	·	·	·	·	·	358	·	·	·	·	·	·	·	·	13	*1000*
CC	**919**	·	·	·	·	·	81	·	·	·	·	·	·	·	·	·	*1000*
Monroe	**1000**	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	*1000*
Expanding	**1000**	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	*1000*
CmteVeto	·	·	·	·	·	·	·	·	·	118	50	30	274	·	1	**527**	*1000*
PlurVeto	·	·	·	·	·	·	·	·	·	**250**	235	173	214	·	·	126	*998*
OMRD	·	·	·	·	·	·	·	·	·	4	1	·	15	·	20	**942**	*982*
SMRD	357	16	9	·	·	·	**469**	14	·	·	·	·	4	·	2	124	*995*
DMRD	354	11	12	·	·	·	**513**	9	1	·	·	·	4	·	2	91	*997*
Random	·	6	3	6	13	2	·	16	7	1	2	·	2	9	3	·	*70*

Table 2 The same data for Exp. 4, with more voters and candidates ($n = 10,001, m = 200$).

In Figure 5, the square that houses the synthetic experiments was shown decomposed into a 3×3 grid, with zones labeled A through I. Table 1 shows, for example, that the most frequent outcome in Experiment 1 for STV elections is 2B/2H, while for Borda it is 1B/3E.

Expanding to many more voters and candidates concentrates the outcomes so that STV is now exclusively giving 2B/2H winners, and Borda is nearly always 4E.

The effects of winner placement on inefficiency ratios are shown in Table 3 and Table 4.

	Exp. 1	Exp. 3	Exp. 4	Exp. 5	Exp. 6
Block	1.42 (3.46)	3.14 (5.88)	3.73 (7.54)	1.30 (2.00)	2.73 (4.47)
Borda	2.15 (3.93)	2.96 (4.71)	4.25 (5.14)	1.48 (1.94)	2.64 (4.46)
STV	1.00 (1.11)	1.00 (1.31)	1.06 (1.17)	1.07 (1.24)	1.00 (1.13)
Ranked Pairs	2.92 (5.28)	3.09 (4.94)	4.98 (6.32)	1.52 (2.01)	2.71 (4.61)
SNTV	1.04 (2.61)	1.00 (3.76)	1.11 (4.89)	1.11 (1.85)	1.00 (1.03)
CC	1.05 (2.62)	1.00 (1.11)	1.12 (4.25)	1.13 (1.54)	1.00 (1.05)
Monroe	1.00 (1.14)	1.00 (1.15)	1.10 (1.24)	1.08 (1.22)	1.00 (1.68)
Expanding Approvals	1.00 (1.09)	1.03 (1.45)	1.01 (1.11)	1.05 (1.25)	1.00 (1.21)
Committee Veto	3.34 (6.04)	4.12 (6.64)	6.49 (7.60)	1.71 (2.37)	3.94 (6.26)
Plurality Veto	2.16 (3.59)	2.89 (4.45)	5.96 (7.02)	1.49 (1.88)	2.58 (4.34)
OMRD	3.73 (6.32)	4.57 (7.68)	7.91 (9.58)	2.33 (3.15)	4.25 (7.27)
SMRD	1.92 (4.50)	3.35 (6.08)	4.11 (7.97)	1.56 (2.54)	2.68 (5.73)
DMRD	1.39 (3.48)	3.05 (5.81)	4.07 (8.02)	1.50 (2.30)	2.26 (5.35)
Random	2.74 (4.85)	3.38 (5.89)	5.05 (7.54)	1.77 (2.53)	2.99 (5.52)

Table 3 Group inefficiency of the worse-off natural bloc is presented as median $\mathcal{I}(\mathcal{B})$ (with 95th percentile score in parentheses) over 1000 trials.

	Exp. 1	Exp. 3	Exp. 4	Exp. 5	Exp. 6
Block	1.11 (1.22)	1.09 (1.20)	1.11 (1.16)	1.10 (1.20)	1.07 (1.16)
Borda	1.01 (1.06)	1.00 (1.03)	1.00 (1.01)	1.00 (1.02)	1.00 (1.02)
STV	1.11 (1.23)	1.16 (1.28)	1.10 (1.14)	1.12 (1.21)	1.15 (1.26)
Ranked Pairs	1.05 (1.15)	1.00 (1.03)	1.01 (1.02)	1.00 (1.02)	1.00 (1.03)
SNTV	1.13 (1.25)	1.19 (1.31)	1.11 (1.15)	1.17 (1.29)	1.18 (1.29)
CC	1.13 (1.24)	1.18 (1.30)	1.11 (1.15)	1.13 (1.21)	1.17 (1.27)
Monroe	1.13 (1.23)	1.18 (1.30)	1.11 (1.15)	1.13 (1.21)	1.14 (1.24)
Expanding Approvals	1.12 (1.23)	1.17 (1.29)	1.11 (1.15)	1.10 (1.18)	1.12 (1.23)
Committee Veto	1.08 (1.21)	1.08 (1.19)	1.02 (1.07)	1.01 (1.07)	1.08 (1.20)
Plurality Veto	1.03 (1.10)	1.00 (1.03)	1.02 (1.05)	1.00 (1.03)	1.00 (1.03)
OMRD	1.16 (1.36)	1.20 (1.39)	1.14 (1.33)	1.18 (1.40)	1.19 (1.37)
SMRD	1.14 (1.30)	1.20 (1.36)	1.14 (1.24)	1.19 (1.36)	1.19 (1.32)
DMRD	1.14 (1.27)	1.19 (1.34)	1.14 (1.24)	1.18 (1.35)	1.19 (1.33)
Random	1.34 (1.58)	1.31 (1.53)	1.42 (1.68)	1.26 (1.47)	1.29 (1.48)

Table 4 Overall inefficiency is presented as median $\mathcal{I}(\mathcal{V})$ (with 95th percentile score in parentheses) over 1000 trials.

E More details on Scottish contests

To extend rules intended for complete ballots to the Scottish STV profiles, unranked candidates are treated as tied at the bottom of each ballot. Randomized rules are run once per contest. Code is available at `https://github.com/mggg/metric-voting`.

Brief description of partial ballot handling:

- **Borda:** position p scores $m - p$, unranked scores 0.
- **Block:** each voter approves their top $\min(k, \text{ballot length})$.
- **STV:** Droop quota, fractional surplus transfer, exhausted ballots drop out.
- **Ranked Pairs:** ranked beats unranked; voters ranking neither abstain.
- **CC, Monroe:** misrepresentation $p - 1$ for position p, $m - 1$ for unranked.
- **Expanding:** a ballot approves only the candidates it ranks.
- **Plurality Veto:** if the bottom choice is unranked, veto a random unranked candidate.
- **Committee Veto:** short top-k sets are padded with random unranked candidates.
- **SMRD, OMRD, DMRD:** an exhausted dictator is replaced by a new random draw.

Tables 5-6 show that the Scottish elections, with fewer candidates and much less separated blocs, have a far higher rate of winner set agreement between voting rules.

	Block	Borda	STV	RkPairs	SNTV	CC	Monroe	Expanding	CmteVeto	PlurVeto	OMRD	SMRD	DMRD
Borda	3												
STV	303	1											
RkPairs	32	90	0										
SNTV	216	0	465	0									
CC	202	0	466	0	798								
Monroe	266	3	704	0	555	554							
Expanding	336	1	700	0	437	454	700						
CmteVeto	41	35	0	391	0	0	0	0					
PlurVeto	2	241	1	45	0	0	1	0	6				
OMRD	25	4	0	64	0	0	0	0	147	0			
SMRD	113	1	126	8	119	107	128	124	19	0	23		
DMRD	120	0	195	1	202	199	198	181	1	0	4	92	
Random	0	0	1	1	1	0	0	0	0	0	0	0	0

Table 5 Number of the 1000 trials of Experiment 1 in which two rules elect the same winner set.

	Block	Borda	STV	RkPairs	SNTV	CC	Monroe	Expanding	CmteVeto	PlurVeto	OMRD	SMRD	DMRD
Borda	.887												
STV	.481	.540											
RkPairs	.895	.908	.512										
SNTV	.363	.418	.724	.389									
CC	.155	.170	.437	.156	.499								
Monroe	.305	.344	.692	.315	.668	.640							
Expanding	.424	.483	.646	.437	.503	.342	.556						
CmteVeto	.229	.223	.116	.238	.098	.025	.075	.112					
PlurVeto	.844	.767	.416	.806	.315	.126	.261	.365	.261				
OMRD	.132	.126	.079	.132	.081	.050	.077	.079	.113	.130			
SMRD	.132	.139	.156	.135	.146	.093	.130	.128	.068	.126	.061		
DMRD	.113	.120	.142	.112	.138	.107	.132	.128	.049	.107	.042	.073	
Random	.015	.014	.014	.017	.020	.025	.017	.014	.017	.015	.020	.020	.025

Table 6 Share of the 717 Scottish contests in which two rules elect the same winner set.

Finally, Figure 10 shows a limited comparison of inefficiency ratios on selected rules and only on the contests where STV and Borda find different winners.

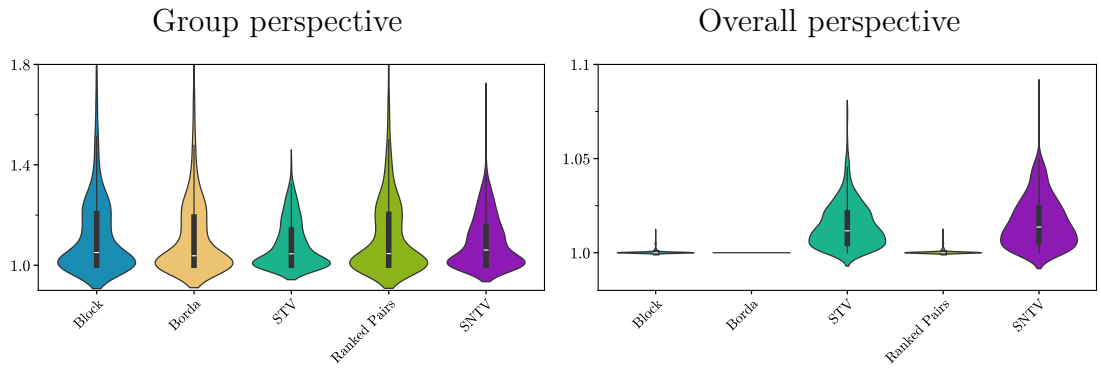


Figure 10 There are 330 contests in the Scottish STV dataset in which there are at least seven candidates and the STV and Borda winner sets are not equal. Over these contests, we find the winners by five voting rules and compare the overall and group inefficiency ratios.